

Aditya Rawat

AI / Computer Vision Engineer

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Portfolio

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SUMMARY

Computer-vision and machine-learning engineer with a newly completed **M.Sc. in Data Science at RWTH Aachen** (thesis result pending). I **fine-tune foundation models for 3D instance segmentation**, benchmark them honestly against the state of the art, and ship the result as tooling people actually use. Equally at home building **autonomous LLM agents** that run unattended in production, with grounding enforced in code rather than requested in a prompt. Two years of prior software-engineering experience; a bias toward reproducibility, measured results, and code that ships.

EDUCATION

M.Sc. Data Science — RWTH Aachen University

10/2023 – 09/2026 (expected)

Coursework complete — grade **2.1**; Master's thesis submitted, degree expected 09/2026. Focus: Computer Vision, Machine Learning, Data Science.

Relevant coursework: Introduction to Artificial Intelligence **1.0** · Machine Learning **1.7** · Computer Science electives **1.4** · Mathematics of Data Science **2.0** · Mathematical Methods of Signal & Image Processing **2.3** · Master Seminar: Current Topics in Computer Vision & ML **2.3** · Advanced Process Mining **1.7** · Business Process Intelligence **1.7**.

B.Sc. Computer Engineering — University of Mumbai

09/2017 – 05/2021

Final grade **8.42 / 10** (First Class with Distinction) · Focus: Software Development, Software Engineering, Machine Learning.

German grading scale: 1.0 = excellent, 4.0 = pass (lower is better).

EXPERIENCE

Student Research Assistant — Computer Vision

07/2024 – 05/2025

Institut für Industrieofenbau und Wärmetechnik (IOB), RWTH Aachen

- Researched **monocular metrology using computer vision** for industrial settings — measuring hot-heel depth in an electric arc furnace.
- Built Python / OpenCV computer-vision tooling and supported PhD researchers with data collection, analysis and software development.

Quality Engineer

06/2021 – 03/2023

Larsen & Toubro Infotech (LTIMindtree), Hyderabad

- Delivered a Python + Selenium **automated-testing framework from proof-of-concept to production, improving testing efficiency by ~50%**.
- Built API test suites (Postman), authored test plans for new features, trained junior engineers, and completed full-stack training (Java / SpringBoot).

SELECTED PROJECTS

Microglomeruli Segmentation — M.Sc. Thesis, RWTH Aachen

11/2025 – 06/2026

3D instance segmentation of synaptic boutons in Drosophila brain microscopy

- **Fine-tuned MicroSAM** (a ViT-based Segment Anything variant, vit_l1m) with custom 3D post-processing for instance segmentation of synaptic boutons in confocal and Airyscan microscopy volumes.
- Hand-built a 3D annotated dataset from scratch in napari — **13 volumetric stacks from 7 brain preparations across two acquisition systems (confocal LSM + Airyscan), split 9 / 2 / 2**; benchmarked head-to-head against nnU-Net v2, CellposeSAM and SwinUNETR under a physically-calibrated, instance-matching protocol.
- The fine-tuned **MicroSAM Large** checkpoint reached **perfect recall (1.000)** and the **top matched-instance mIoU (0.787)** of every model evaluated — ahead of all three SOTA backbones and the lab's semi-manual Imaris baseline; checkpoints published on Hugging Face.
- Shipped **BoutonViewer**, a napari desktop app that runs the pipeline and reports per-bouton volume and surface area in μm^3 / μm^2 for non-programmer biologists.

Stack: PyTorch · MicroSAM · MONAI · napari · Python · HTCondor

TechDrishti — Agentic AI for autonomous journalism

2026

- Built a **fully autonomous, multi-stage agentic LLM pipeline** that collects, triages, researches and writes original Hindi tech-news articles daily — **unattended on GitHub Actions at ~1¢ per run, zero servers**.
- Engineered cheap-model triage, a persistent entity cache with sense disambiguation, and sentence-embedding clustering to deduplicate coverage across 8+ feeds.

Stack: Python · LLM orchestration · sentence-transformers · GitHub Actions

- ReAct agent that plans its own searches and **enforces grounding in code, not prompts**: a code-level check flags any comparison claim whose target was not found in retrieved sources, plus an AST-based calculator and a keyless retrieval layer. Live on Hugging Face Spaces.
Stack: Python · ReAct / tool use · Chainlit · FastAPI

Chitragupt — Vision-language agent

2026 · in progress

- Agentic assistant on a **vision-language model** that reasons over images and calls tools; VLM served on a free Colab GPU via a FastAPI bridge to web, desktop and CLI clients.
Stack: VLM · FastAPI · Python · Computer Vision

Celonis Cohort Discovery & Analysis Web App — Interdisciplinary Lab, w/ Celonis

04/2024 – 07/2025 · Grade 1.7

- Implemented a research-paper algorithm to parse and segment process/event logs into cohorts, and built the **full-stack app (React + Flask + PyTest)** for filtering, visualization and comparative cohort analysis. Agile / TDD.
Stack: React · Flask · Python · Celonis · process mining

Realtime Human Detection & Counting — Bachelor's project

10/2020 – 05/2021

- **Led a team of 3** to build a real-time human-detection and social-distancing system (**YOLOv3** + perspective transformation) for COVID-19 protocol enforcement.
Stack: YOLOv3 · OpenCV · Python

SKILLS & TOOLS

Programming	Python · C++ · Java · TypeScript · SQL
Computer Vision	PyTorch · Transformers · Vision Transformers (ViT) · MicroSAM / SAM · nnU-Net v2 · Cellpose · SwinUNETR · StarDist · MONAI · napari · OpenCV · YOLOv3 · 3D instance segmentation · foundation-model fine-tuning
LLM & Agentic AI	LLM pipeline design · ReAct agents · tool use / function calling · vision-language models (VLMs) · grounding & hallucination mitigation · sentence-transformers · evals
ML & Data	scikit-learn · Pandas · NumPy · Matplotlib / Seaborn · process mining (Celonis, ProM)
MLOps & Tooling	Git · Docker · Hugging Face · GitHub Actions · HTCondor clusters · FastAPI · Flask · React · Chainlit · PyTest · Postman · Linux

LANGUAGES

Hindi (native) · English (C1, TOEFL — near-native) · German (A2, in progress)